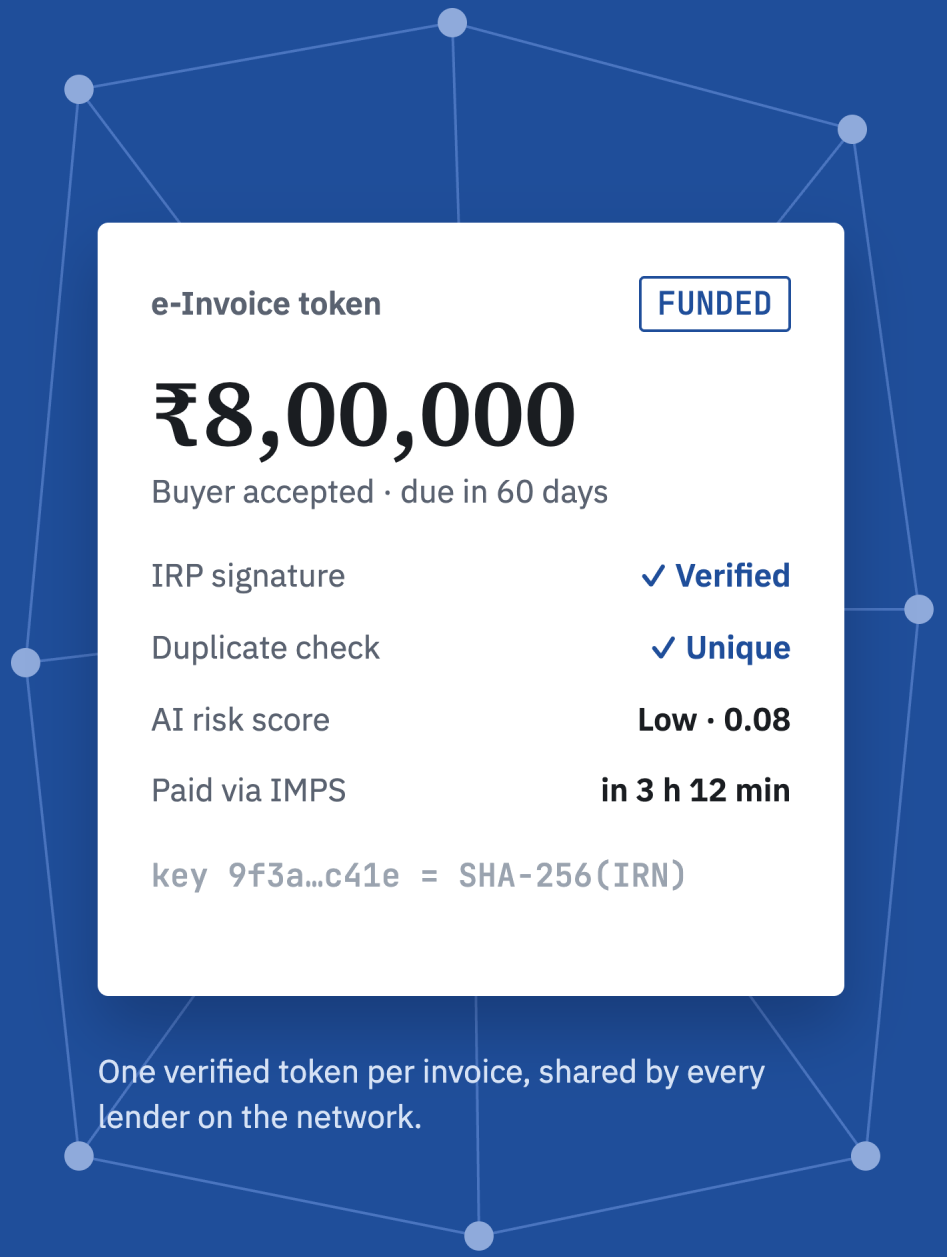
Citi × NPCI DRUNIX Hackathon 2026 · Stage 3 idea pitch

RASID

Hindi for “receipt”

Fraud-proof invoice finance for India’s MSMEs, built on DRUNIX and paid in hours, not months.

Team	Registration ID	Primary theme
ARD	**STU5011**	**Real Asset Tokenization**



MSMEs wait months to be paid, and lenders can't safely fund the wait

₹10.7 lakh cr

owed to MSMEs in delayed payments[1]

60–120 days

typical wait to be paid, despite the 45-day legal limit[2]

₹20–25 lakh cr

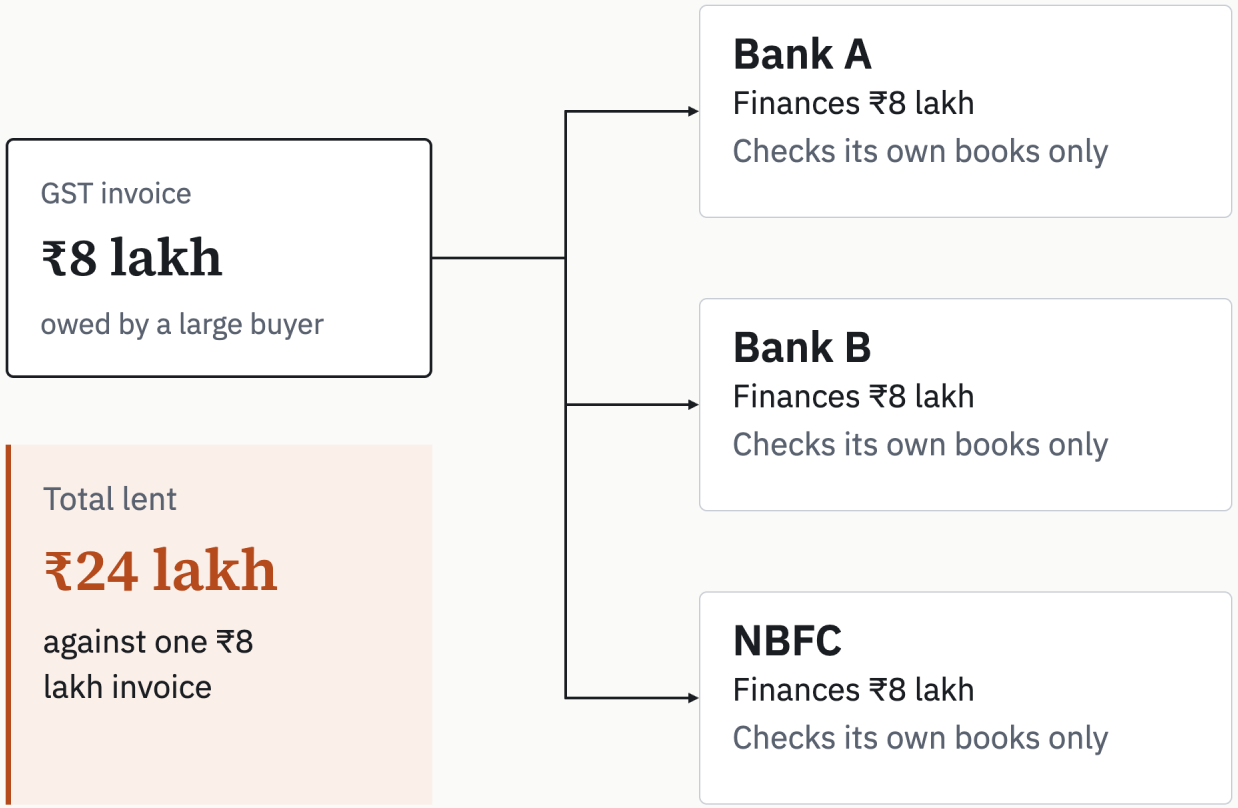
estimated formal MSME credit gap[3]

Who carries the cost

Group	What happens today	What it costs them
MSME suppliers	Paid in 60–120 days, no collateral to borrow against	Informal credit at 18–30% a year
Banks and NBFCs	Can't see what other lenders have financed	Fraud losses, slow manual checks
Large buyers	Separate supplier-finance scheme with each bank	Duplicate onboarding, 43B(h) exposure
TReDS platforms	Duplicate checks stay inside one platform	Lenders hesitate to bid

Sources: [1] GAME report on delayed payments (2022). [2] MSMED Act s.15; Income Tax Act s.43B(h). [3] IFC and SIDBI estimates.

The root cause: every lender checks only its own books



Four gaps in today's system

1. Fragmented registries

TReDS platforms, bank portals and NBFCs check duplicates only internally.

2. One-time checks

An invoice is verified at onboarding; nobody tracks it after that.

3. Rules miss patterns

Circular trading and slightly altered duplicates pass static rules.

4. Manual settlement

Payment references are matched to invoices and split by hand.

RASID gives every invoice one verified, tradable record on DRUNIX

1

Verify

Checks the IRP-signed QR inside chaincode.

JWT verified against IRP keys; no oracle.

2

Register once

One token per invoice, across all lenders.

Key = SHA-256(IRN); MVCC stops races.

3

Finance privately

Sealed bids; the MSME takes the best rate.

Bids go through KeyDB; only hashes on-chain.

4

Settle automatically

Paid in hours; repaid and split on the due date.

Payment UTRs confirm each state change.

What changes for each user

User	Today	With RASID
MSME	Waits 60–120 days; borrows at 18–30%	Paid within hours at buyer-risk rates
Lender	Blind to other lenders’ exposure	Network-wide duplicate check and AI score
Buyer	Separate scheme with each bank	One acceptance works with every lender
NPCI	No shared rail for receivables	Neutral network for tokenized invoices

One invoice, from upload to repayment, in five steps

Step 1

Upload

MSME

Photo, voice or e-invoice file, in any Indian language

`RegisterInvoice`

REGISTERED · minutes

Step 2

Accept

Buyer

Confirms amount and due date in one click

`AcceptInvoice`

ACCEPTED · same day

Step 3

Bid

Lenders

Sealed bids during a fixed bid window

`SubmitBid`

LISTED · bid window

Step 4

Get paid

MSME

Best bid wins; money arrives by IMPS or UPI

`ConfirmFunding`

FUNDED · within hours

Step 5

Repay

Buyer

e-NACH debit on the due date; lenders paid pro rata

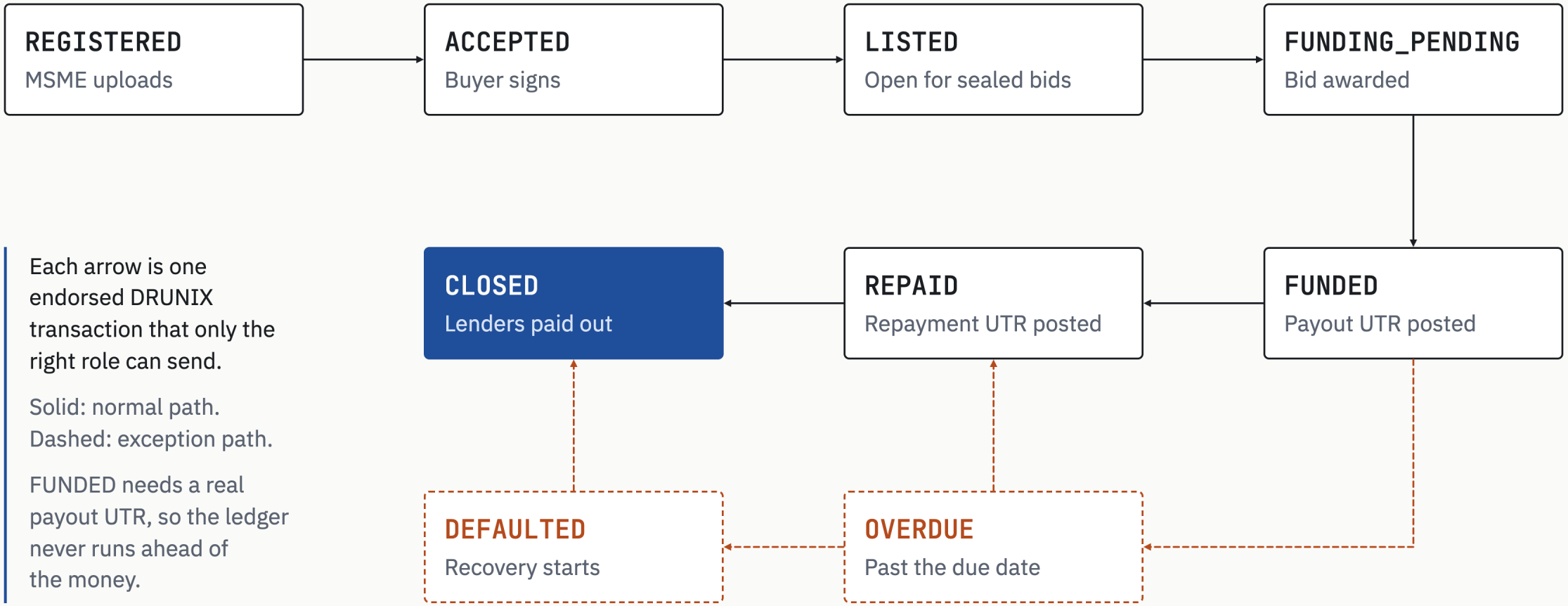
`ConfirmRepayment`

CLOSED · due date

Time from delivery to cash



Each invoice token follows one path, enforced in chaincode

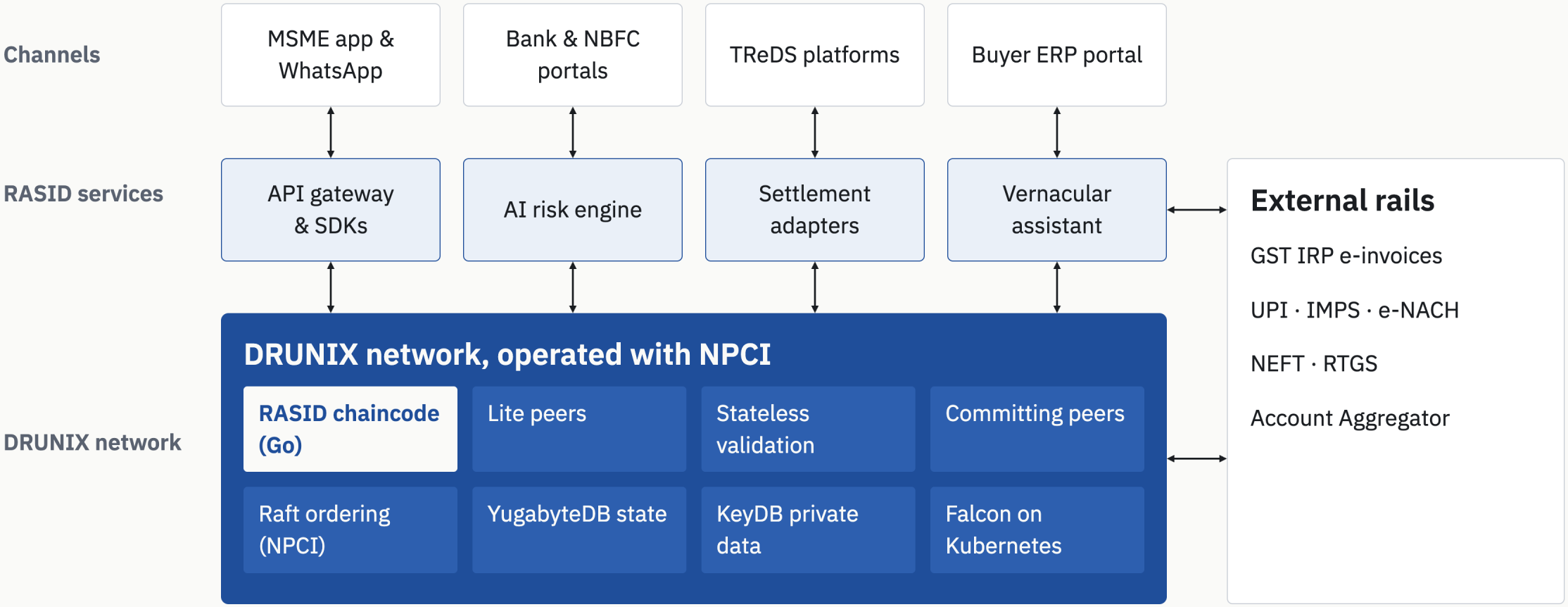


Each arrow is one endorsed DRUNIX transaction that only the right role can send.

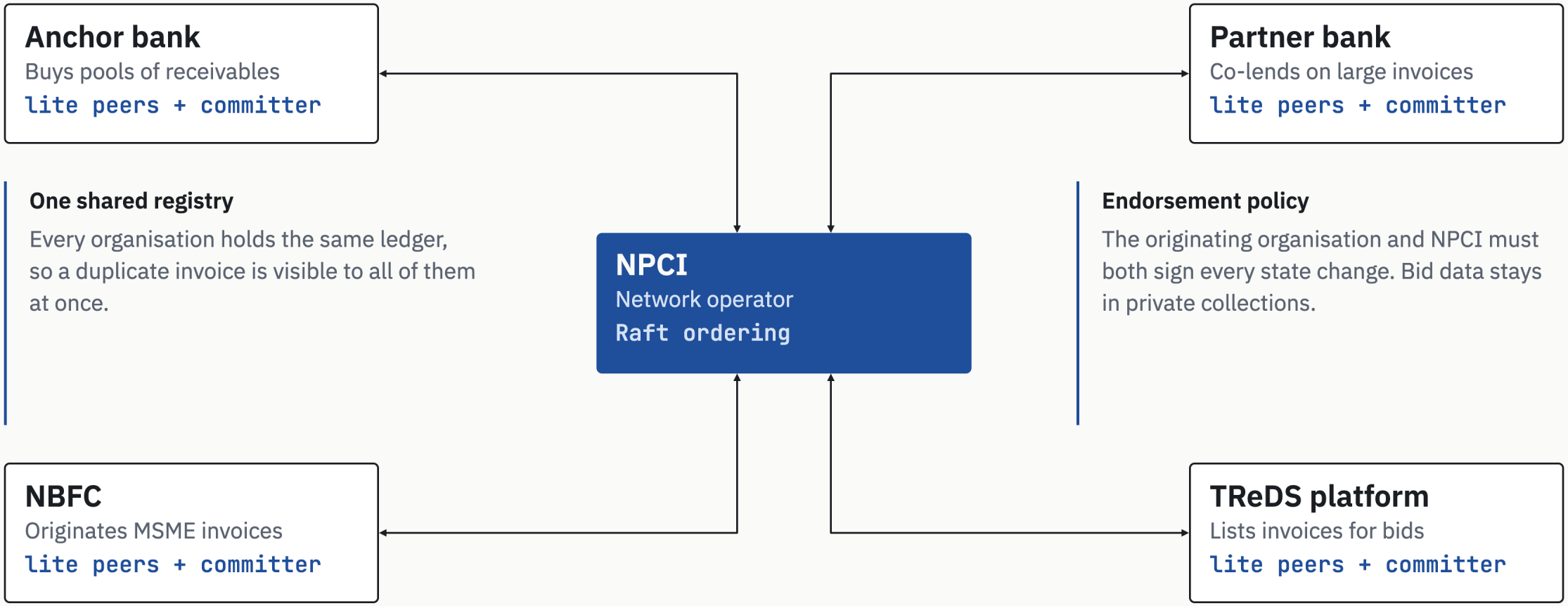
Solid: normal path.
Dashed: exception path.

FUNDED needs a real payout UTR, so the ledger never runs ahead of the money.

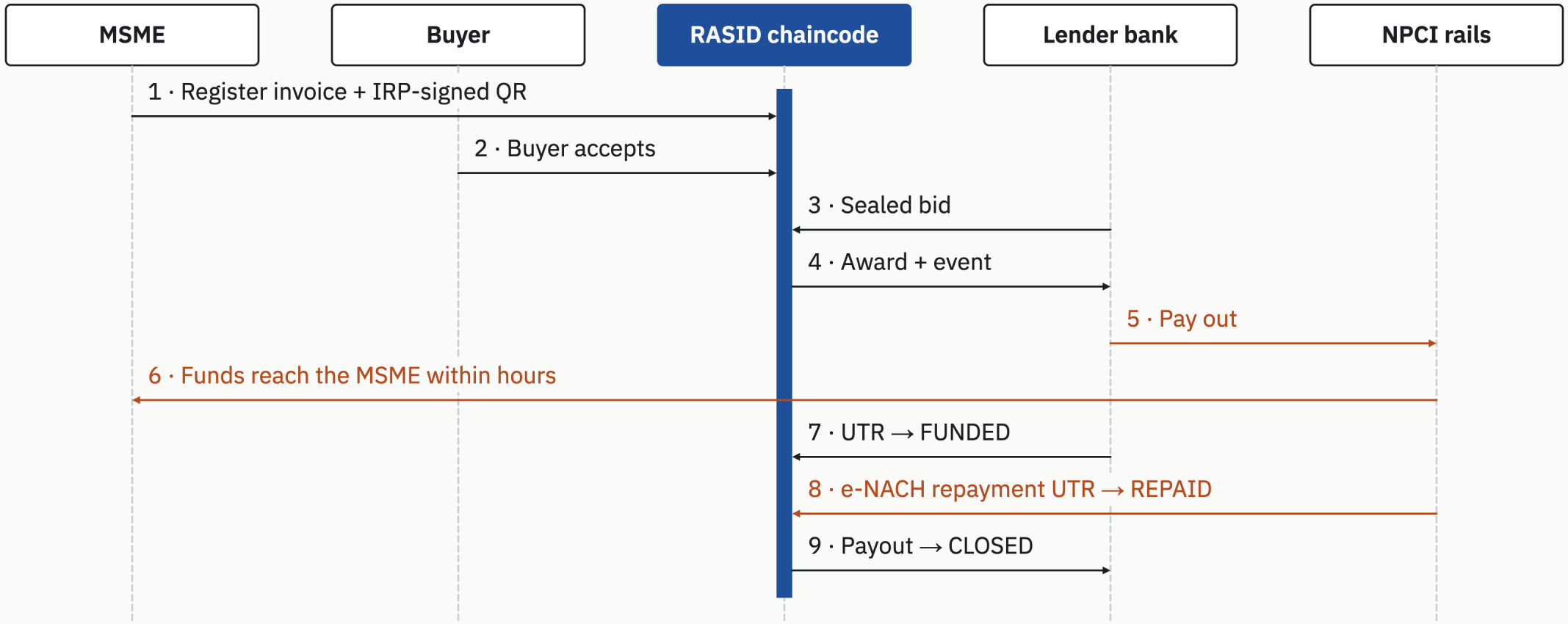
Users, RASID services and payment rails all meet on DRUNIX



Every lender runs its own peers; NPCI orders and co-endorses

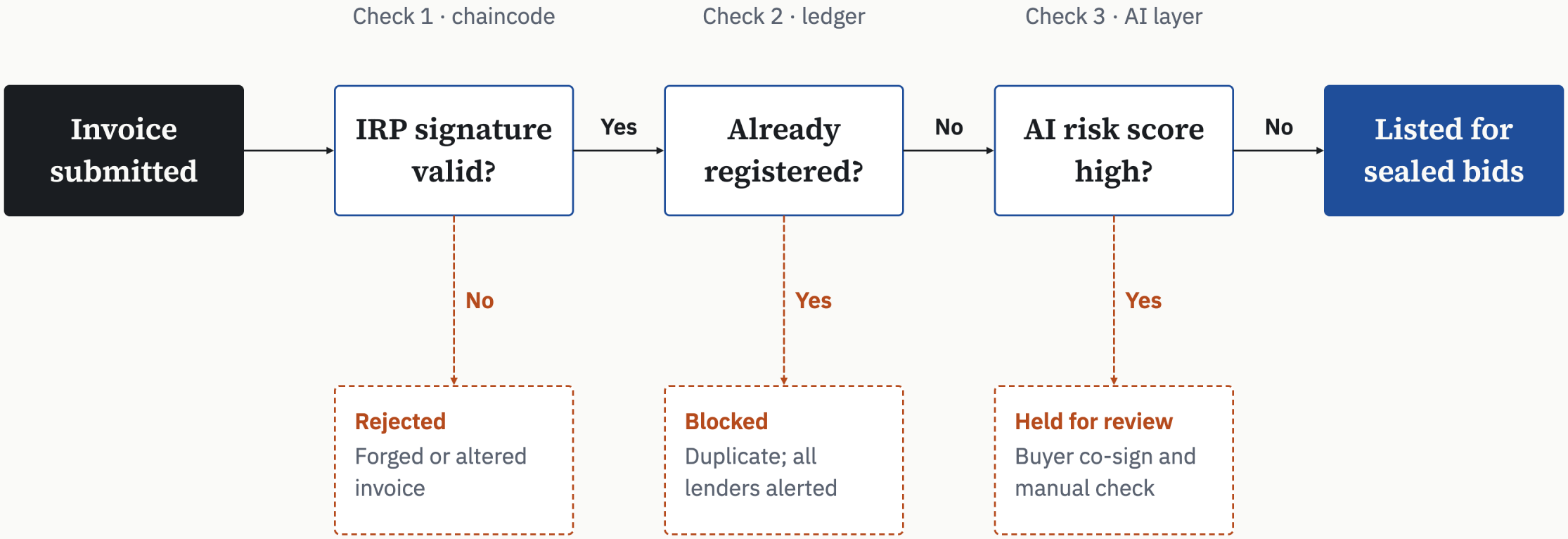


One invoice end to end: nine steps across DRUNIX and NPCI rails



Dark arrows are DRUNIX transactions. Orange arrows are money moving on NPCI rails.

Three checks stop forged, duplicate and risky invoices before listing



Four Go chaincode contracts hold all of RASID's business rules

Contract	Key functions	What it enforces
InvoiceRegistry	`RegisterInvoice, AcceptInvoice, MarkOverdue`	IRP signature, unique key, right role per step
Marketplace	`OpenAuction, SubmitBid, AwardBid`	Sealed bids, bid window, best-rate award
ReceivableToken	`SplitUnits, TransferUnits, BalanceOf`	Units always sum to the invoice amount
Settlement	`ConfirmFunding, ConfirmRepayment, Distribute`	One UTR per payment, pro-rata payouts

Token record (shared ledger)

`key = SHA-256(IRN) · state · amount · dueDate`

`sellerHash · buyerHash · units[] · holders[]`

`riskScoreHash · modelVersion`

`fundingUTR · repaymentUTR · auditTrail`

Private data (KeyDB and collections)

`bidRate · bidAmount · bidderOrg`

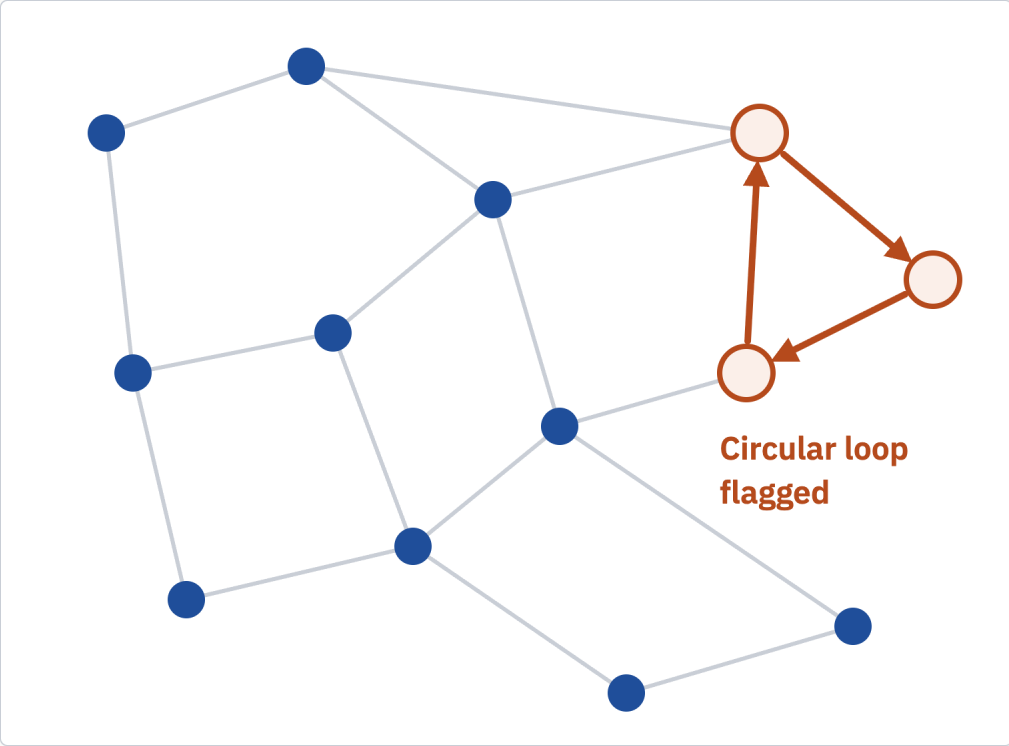
`msmeBankAccount · contactDetails`

`aaCashFlowSummary (consented)`

Purged after blockToLive, for DPDP compliance

Every contract runs under one endorsement policy: the originating organisation and NPCI must both sign.

The AI layer sees the whole invoice network, which no single lender can



Each dot is a firm; each line is an invoice between two firms.

Circular trading

Graph analysis across all lenders finds loops where related firms bill each other to create fake receivables.

Near-duplicates

Fuzzy matching on amount, date, line items and GSTINs catches re-issued invoices that pass the exact-key check.

Buyer behaviour

On-ledger repayment history and consented Account Aggregator data, so lenders price on the buyer's risk.

Audit trail

Each score is signed, and its hash and model version are written to the token, so any decision can be audited.

> Prototype: trained on synthetic invoice networks with injected fraud patterns.

How RASID uses each DRUNIX building block

DRUNIX element	How RASID uses it	Why it matters
Lite + committing peers	Lite peers scale out; few committers per org	Absorbs month-end spikes cheaply
Stateless validation	Checks signatures and policy in its own service	Stays fast while commits continue
YugabyteDB state	Token state with SQL for exposure reports	LevelDB and CouchDB lack SQL at scale
KeyDB + private data	Bids and bank details stay off the shared ledger	Pricing stays private but provable
Chaincode (Go)	Registry, auction, token and settlement contracts	All business rules in one audited place
Multi-org network	Banks, NBFCs, TReDS and NPCI as separate orgs	No single lender controls the registry
Falcon	Helm charts for each org's peers on Kubernetes	Repeatable onboarding for new banks
APIs and integrations	UPI, IMPS, e-NACH, GST IRP, AA, Caliper	Real money and identity, no new rails

Planned contributions back to DRUNIX (Apache 2.0)

Caliper benchmark pack
Invoice workloads to test lite-peer scaling

Sealed-bid auction module
Reusable chaincode built on private data

Falcon autoscaling profiles
KEDA rules that add lite peers under load

At 5% of the delayed-payment pool, RASID saves about ₹790 crore a year

Assumptions (year 3)

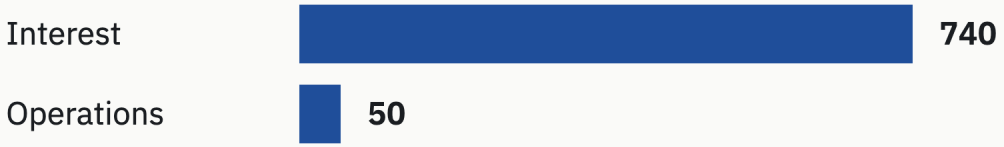
Input	Value
Volume financed	₹50,000 cr, 5% of ₹10.7 lakh cr
Average invoice	₹2 lakh, so 25 lakh invoices
MSME cost of credit	19% informal → 10% with RASID
Average tenor	60 days
Ops saving per invoice	₹200 of manual checks
Financed per MSME	₹1 cr a year, so 50,000 MSMEs

19% is the conservative end of the 18–30% informal range.

Cost of credit for an MSME



Annual savings, ₹ crore




```
₹50,000 cr × 9% × 60/365 ≈ ₹740 cr
25 lakh invoices × ₹200 ≈ ₹50 cr
```

Savings scale with volume: about ₹316 cr at 2% of the pool and ₹1,580 cr at 10%. Time to cash falls by over 95%.

RASID adds what today's invoice-finance options lack

Capability	Typical today	RASID on DRUNIX
Duplicate check	Each lender checks its own books	One registry across all lenders
Invoice authenticity	PDF upload and manual review	IRP signature verified in chaincode
Lender pricing	Bilateral deals or open quotes	Sealed bids, provable by hash
Settlement	Manual reconciliation, days	UTR-confirmed, within hours
Fraud detection	One lender's data only	AI on the network-wide invoice graph
Neutrality	Platform owned by one party	Run with NPCI; no member controls it

Why it is hard to copy

Network effect
Each new lender makes double financing harder for every member.

Neutral operator
A shared ledger run with NPCI is something no single lender can build.

Open contributions
Our DRUNIX modules keep RASID aligned with the platform's roadmap.

Key risks and how we handle them

Risk	Why it matters	Mitigation
Regulation	Tokenized receivables are new for regulators	Work within TReDS and factoring rules; apply to the RBI regulatory sandbox
Adoption	Value grows only as lenders join	Pilot with one anchor bank and its NBFC partners; buyers accept once for all lenders
Data privacy	GST and bank data are personal data under DPDP	Only hashes on-chain; private collections purged on schedule; consent via Account Aggregator
Throughput	Month-end GST filing creates sharp peaks	Lite peers autoscale; Caliper benchmarks before any pilot
Model risk	A false flag could block an honest MSME	AI only holds invoices for review, never rejects; scores signed and versioned
Buyer default	Lenders lose if the buyer does not pay	e-NACH mandate at acceptance; priced on buyer risk; OVERDUE and DEFAULTED paths on-chain

Plan: prototype by 22 November, pilot with one bank in 2027

25 Oct — Pitch
Proposal and this deck submitted

10–22 Nov — Build
Network, contracts, AI model and demo app

22 Nov — Ship
Code and README in rasid-drunix

4 Dec — Finale
Live demo of one invoice, end to end

2027 — Pilot
One anchor bank and its NBFC partners

Prototype scope

Three-organisation DRUNIX network deployed with Falcon

Four Go contracts with unit tests

Mock IRP signer and UPI/IMPS sandbox adapters

Fraud model on synthetic invoice graphs

Caliper benchmark of lite-peer scaling

Budget: ₹3,00,000 over 45 days



- Cloud and Kubernetes · ₹1.2 L
- AI model and assistant · ₹50 k
- Integrations · ₹30 k
- Security · ₹30 k
- Data and design · ₹20 k
- Contingency · ₹50 k

RASID

The trust layer for India’s MSME receivables, built on DRUNIX.

₹790 cr
saved a year at 5% of the pool

50,000
MSMEs reached

Hours
to cash, not months

`github.com/AradhyaSomani/rasid-drunix`

Team ARD · Registration ID STU5011